

Advancing the INFORM Hazard Evidence for Tanzania while Preserving the INFORM Engine: a CHIRPS/ERA5/NBS multi-criteria drought case study

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Status - proposal for INFORM Calc Engine v2. The operational model (**v1**) keeps the INFORM aggregation engine **exactly** as the official Excel workbook - verified formula-by-formula, written in the code in the workbook's own literal form (product GEOMEAN, product of cube roots, AVERAGE, min-max). The advancements documented in this paper are a **proposal**; once validated ("when it passes") they are implemented as **INFORM Calc Engine v2**. In short: **v1 = exact INFORM now; v2 = these advancements next**. This is one of three documents - (1) the operational **Manual**, (2) the **Exact-formula reference** (INFORM_CALCULATION_ENGINE.docx, what v1 runs), and (3) this **Paper** (the v2 proposal).

Abstract

The INFORM Subnational Risk Index aggregates standardised (0-10) hazard, vulnerability and lack-of-coping indicators into a comparable risk score. Its aggregation **engine** - min-max standardisation (with logarithmic transformation and Tukey outlier capping), arithmetic averaging of indicators into categories, the **scaled geometric mean** of categories into dimensions, and the **cube root** of the three dimensions into risk - is a fixed methodology we **preserve exactly** as implemented in the official *Tanzania - Country Model Template* workbook (verified formula-by-formula from its hidden sheets). What we **advance** is the *evidence* behind individual hazard indicators where authoritative national data exist. This paper documents the advancement of the **drought hazard** from a coarse INFORM SADC value to a transparent four-component index computed from CHIRPS v3 rainfall (1991-2024) and ERA5 temperature - aridity, interannual variability, SPEI drought depth, and growing-season failure - together with the **initial vs advanced** comparison, the **impact**, and the **gap** the advancement closes. We show the initial drought layer was spatially incoherent (the central semi-arid zone under-rated at ~4/10, a coastal city over-rated at 10/10), whereas the advanced layer tracks observed rainfall and

the food-security geography. The flood, heatwave and exposure advancements are summarised; engine-level refinements (IQR capping of recomputed indicators, custom vs data-range references) are deferred to future work by design.

1. Introduction

INFORM (the Index for Risk Management; DRMKC/JRC, European Commission) provides a transparent, reproducible composite risk model [1]. For Tanzania, the operational instrument is the *Tanzania - Country Model Template* workbook, whose engine standardises each indicator to 0-10 and aggregates them. The engine is deliberately conservative and globally comparable; it must not be silently altered.

Our work separates two concerns:

1. **The engine** (how numbers combine). This stays **identical** to the workbook - we verified every formula directly from the (hidden) engine sheets (see the companion `INFORM_CALCULATION_ENGINE.docx`): component = AVERAGE, category = AVERAGE, dimension = $\text{ROUND}((10 - \text{GEOMEAN}((10 - c)/10^{*9+1} \dots))/9 * 10, 1)$ (scaled geometric mean), risk = $\text{ROUND}(H^{(1/3)} * V^{(1/3)} * LCC^{(1/3)}, 1)$ (cube root), and standardisation = min-max to 0-10 with optional log/exp transform, Tukey (Q1-1.5IQR, Q3+1.5IQR) outlier capping, and an inversion for “Decrease Risk” indicators.
2. **The evidence** (the value of each indicator). Where the INFORM SADC layer is coarse, dated, or incoherent for Tanzania, and authoritative national data exist, we **recompute the indicator** from primary data using the **same standardisation method** the engine prescribes, and we apply it **raise-only** (the computed value may raise a documented hazard but never silently lower it).

This paper makes the second concern explicit and auditable, using **drought** as the worked case, because drought is the hazard most consequential for Tanzanian food security and the one for which the densest observational record exists.

2. The preserved INFORM engine

We reproduced a unit end-to-end against the workbook: for Kondoia, the workbook’s own Hazard dimension is $\text{scaled-geomean}(\text{Natural } 3.544, \text{Human } 0.683) = 2.2$, and our implementation returns the same 2.2 for the same inputs; the national value matches the workbook to the decimal (Hazard 2.2, Vulnerability 5.5, Lack-of-Coping 5.9, Risk 4.1). The engine is therefore **unchanged**; only indicator *inputs* differ where we advanced them. Two engine-level options the workbook supports but our recompute scripts do not yet apply - **Tukey IQR capping** of recomputed series, and **custom (fixed) reference ranges** instead of the Tanzania data range - are recorded as **future work** (Section 7) and are not changed here.

3. The drought advancement - method

3.1 Rationale and the gap in the initial layer

The initial INFORM SADC drought scores for Tanzania are spatially incoherent (Section 5, Table 2): several central **semi-arid** councils (Kondoa 771 mm yr⁻¹, Singida 749 mm, Manyoni 760 mm) scored only 3.6-4.3/10, while a wet coastal city (Dar es Salaam / Ilala, 1,253 mm) scored 10/10. Semi-arid central Tanzania is a recurrent drought- and food-insecurity hotspot (IPC/MUCHALI classifications), so an index that rates it below 5 while maxing out a coastal city is not credible. This is the **gap**: the *initial* relevance of the drought layer was, in places, simply not true.

3.2 Data

- **CHIRPS v3** monthly precipitation, 0.05 deg, 1991-2024, the same product validated for Tanzanian rainfall classification [2]. Zonal-mean rainfall is extracted per administrative unit.
- **CHIRPS v3 daily** (2015-2024) for the companion heavy-rain/flood layer (Section 6).
- **ERA5** 2 m temperature, used to estimate potential evapotranspiration (PET) for SPEI [3].

3.3 The four-component index and its criteria

Drought relevant to communities is more than a single statistic; following the drought-index literature [4,5] we combine four physically distinct, individually justified components, each min-max scaled to 0-1:

Component	Weight	What it captures	Computation	Basis
Aridity	0.30	inherent dryness of the climate	min-max of (-mean annual rainfall)	UNEP aridity gradient; semi-arid = ~0.2-0.5 P/PET [6,7]
Interannual variability	0.20	year-to-year rainfall instability	min-max of the coefficient of variation of annual rainfall	East-African rainfall is among the world's most variable [8,9]
SPEI drought depth	0.15	severity of water-balance deficits	mean magnitude of SPEI-12 < -1 , PET via Thornthwaite [10] from ERA5	SPEI = standardised precipitation-evapotranspiration index [3,11]
Growing-season failure	0.35	crop-failure / food-security driver	0.5CV + 0.5frequency of the main 4-month season below 80% of normal	season failure -> IPC food crises (highest weight)

The blended score $0.30 \times \text{aridity} + 0.20 \times \text{variability} + 0.15 \times \text{SPEI-depth} + 0.35 \times \text{season-failure}$ is then **min-max standardised to 0-10** - exactly the INFORM normalisation - and applied **raise-only** (the advanced value replaces the initial only where it is higher, so no documented drought area is downgraded).

Weighting rationale. Aridity is the baseline (a dry climate is intrinsically drought-exposed); variability and SPEI add the *dynamic* deficit; **growing-season failure carries the largest weight (0.35)** because, for an operational risk index feeding humanitarian planning, the decision-relevant outcome is failure of the main cropping season, which is what triggers IPC/MUCHALI food-insecurity phases. The weights are an explicit modelling choice; a sensitivity analysis is listed as future work.

Why SPI/SPEI frequency is not used to rank. Drought *frequency* (the share of months below a threshold) is approximately stationary by construction across a fixed record and therefore discriminates poorly between districts; we use SPEI **depth** (how deep deficits go) rather than frequency, and rank on aridity + variability + season-failure, which are the spatially-informative signals.

4. Standardisation and integration (unchanged engine)

The four raw components are combined and the result is min-max scaled to 0-10 using the **same** formula the workbook applies to every indicator ($10 \times (x - \text{Min}) / (\text{Max} - \text{Min})$), clamped, rounded to 0.1). The drought score then enters the **unchanged** chain: it averages with the other natural-hazard components into the *Natural* category, which scaled-geometric-means with *Human* hazards into the *Hazard & Exposure* dimension, which cube-roots with Vulnerability and Lack-of-Coping into Risk. No engine constant is altered.

5. Results - initial vs advanced

Table 2. Initial (INFORM SADC) vs advanced (CHIRPS/ERA5) drought score, with the raw evidence.

District	Rainfall (mm yr ⁻¹)	Rain CV	Season- fail freq	SPEI depth	Initial	Advanced
Longido (pastoral)	668	0.26	0.27	1.60	8.6	9.7
Monduli (pastoral)	694	0.26	0.27	1.56	8.6	9.2
Singida (semi- arid)	749	0.24	0.24	1.68	3.6	8.7
Manyoni (semi-	760	0.20	0.18	1.63	3.6	7.0

District	Rainfall (mm yr ⁻¹)	Rain CV	Season- fail freq	SPEI depth	Initial	Advanced
arid)						
Kondoa (semi- arid)	771	0.24	0.24	1.64	4.3	8.6
Rufiji (sub- humid)	1,088	0.21	0.21	1.50	2.9	5.8
Kilombero (wet)	1,598	0.17	0.21	1.45	0.7	2.9
Kyela (wettest)	1,996	0.16	0.21	1.48	0.0	2.2
Ilala / Dar (coastal city)	1,253	0.26	0.24	1.49	10	10*

*Dar es Salaam retains 10 under the raise-only floor; the computed value is ~6-7. This is a flagged artifact (the initial 10 is documented-but-wrong) queued for correction via Data Entry, *not* an advancement outcome.

Pattern. The advanced layer is **monotone in rainfall**: the five driest councils (Longido, Monduli, Singida, Manyoni, Kondoa, all 668-771 mm) score 7.0-9.7; the wettest (Kyela, Kilombero, 1,600-2,000 mm) score 2.2-2.9. The initial layer was not monotone - it under-rated Singida/Manyoni/Kondoa (3.6-4.3) despite their identical semi-arid climate to the (correctly high) pastoral districts.

6. Other advancements (summary)

The same *initial* -> *advanced* (same standardisation) -> *raise-only* discipline was applied to:

- **Flood / heavy rainfall.** Advanced from a static INFORM value to **CHIRPS v3 daily** counts of days > 50 mm and > 100 mm (2015-2024), combined with documented flood events and the INFORM **Hazard x Exposure** form $\text{flood} = \max(\text{hazard}, \sqrt{\text{hazard} * \text{exposure}})$ so population exposure can only amplify a flood hazard, never hide it. Heavy-precipitation thresholds follow the ETCCDI extreme-rainfall indices.
- **Heatwave.** Advanced from blank/baseline to an **ERA5** temperature-climatology exposure proxy (mean + hottest-quarter), min-max capped - hot lowlands high, cool highlands low.
- **Exposure.** Advanced to **NBS 2022 PHC** population density (population / area), **log** min-max standardised - the workbook's own "Logarithm" transformation for the right-skewed density distribution.

Vulnerability (poverty, stunting, HIV+malaria, displacement, food insecurity) and parts of coping (WASH, health/education access, DRR) were similarly refreshed from NBS/TDHS/IPC/UNHCR sources. All used the engine's standardisation unchanged.

7. Discussion: initial vs advanced, impact, and the gap

Initial vs advanced. The advancement does not change *how* drought enters risk; it changes *what value* drought has, from a coarse/incoherent layer to one grounded in 34 years of observed rainfall and the water balance.

Impact / outcome. Risk rises where the evidence supports it (the central semi-arid belt and the pastoral north), in line with IPC/MUCHALI food-insecurity classifications, and the change is auditably raise-only. Because risk is a cube root of three dimensions, a drought rise of several points lifts the Hazard dimension modestly and the risk score by a fraction - appropriately, since drought is one of twelve natural hazards.

The gap. The advancement is justified precisely where **the initial relevance was not true**: Kondo and Singida are textbook semi-arid, food-insecure districts that the initial layer scored below 5. The advancement closes that gap with transparent, reproducible evidence. Where the initial layer was already correct (pastoral Longido/Monduli), the advancement merely confirms and slightly raises it; where the initial layer was wrong-high (the Dar artifact), raise-only deliberately does **not** advance it down - that case is handled separately as a documented correction.

8. Limitations and future work (engine stays as INFORM)

- **Engine refinements deferred:** apply the workbook's **Tukey IQR capping** to our recomputed series, and decide per indicator between **Tanzania data-range** and **INFORM custom (fixed global/SADC) references**. These change absolute 0-10 values and so are reserved for a future, separately-validated revision.
- **Drought exposure:** drought is currently a pure hazard; pairing it with **agricultural exposure** (cropland/livestock, not urban population) is future work.
- **Weights:** the 0.30/0.20/0.15/0.35 blend is an explicit choice; a sensitivity analysis is planned.
- **Wildfire, lightning, storms** remain documented overlays pending physical computation (MODIS/VIIRS, flash-density, cyclone tracks); the USGS catalogue was used only to **cross-check** (not replace) the earthquake overlay.

References

[1] INFORM (2024). *INFORM Subnational Risk Index - SADC*. Disaster Risk Management Knowledge Centre / Joint Research Centre, European Commission. [2] Funk, C. et al. (2015). The climate hazards infrared precipitation with stations - a new environmental record for monitoring extremes. *Scientific Data* 2, 150066. [3] Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999-2049. [4] Zargar, A. et al. (2011). A review of drought indices. *Environmental Reviews* 19, 333-349. [5] WMO (2012). *Standardized*

Precipitation Index User Guide (WMO-No. 1090). [6] UNEP (1992). *World Atlas of Desertification*. United Nations Environment Programme. [7] Middleton, N. & Thomas, D. (1997). *World Atlas of Desertification*, 2nd ed. UNEP. [8] Nicholson, S. E. (2017). Climate and climatic variability of rainfall over eastern Africa. *Reviews of Geophysics* 55, 590-635. [9] Lyon, B. & DeWitt, D. G. (2012). A recent and abrupt decline in the East African long rains. *Geophysical Research Letters* 39, L02702. [10] Thornthwaite, C. W. (1948). An approach toward a rational classification of climate. *Geographical Review* 38, 55-94. [11] Vicente-Serrano, S. M., Begueria, S. & Lopez-Moreno, J. I. (2010). A multiscalar drought index sensitive to global warming: the SPEI. *Journal of Climate* 23, 1696-1718. [12] McKee, T. B., Doesken, N. J. & Kleist, J. (1993). The relationship of drought frequency and duration to time scales. *Proc. 8th Conf. on Applied Climatology*, 179-184.